MIL-A-28844A(SH)
19 December 1986
SUPERSEDING
MIL-A-28844(EC)
15 August 1978
(See 6.5)

MILITARY SPECIFICATION

ANTIWETTING COMPOUND 3-HEPT

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for technical grade antiwetting compound 3-(heptafluoroisopropoxy) propyltrichlorosilane (3-HEPT).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

PPP-B-636 - Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6850

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 133 - Standard Specification for Distillation Equipment.
E 260 - Standard Recommended Practice for General Gas Chromatography Procedures.
E 355 - Standard Recommended Practice for Gas Chromatography Terms and Relationships.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNITED STATES PHARMACOPEIA

Chemical Resistance, Glass Containers Type I

(Application for copies should be addressed to the U.S. Pharmacopeia Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order or precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical requirements. The antiwetting compound 3-HEPT shall conform to the requirements specified in table I when tested as specified in 4.4.2.

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TABLE I. Requirements.

Characteristic	Requirement	
	Minimum	Maximum
Silane content (as 3-HEPT), percent by weight	99.5	----
Distillation range at 35 millimeters (mm) (1.38 inches) of mercury (Hg)		
Initial boiling point, degrees Celsius (°C)	85	----
Dry point, °C	----	87
Iron content, parts per million (p/m)	----	2.0

3.2 Qualitative requirements.

3.2.1 Appearance. The antiwetting compound 3-HEPT shall be a clear, transparent, colorless liquid, and shall not contain visible insoluble material.

3.3 Material safety data sheet (MSDS). The contracting activity shall be furnished a MSDS at the time of contract award. The MSDS is form OSHA-20, found as part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

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4.2 Sampling for inspection. Unless otherwise specified (see 6.2), sampling for inspection shall be performed in accordance with MIL-STD-105. For purposes of sampling, an inspection lot for examinations and tests shall consist of all the products made from the same raw materials by the same methods and submitted for inspection at the same time.

4.3 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels for determining the sample size, and the AQLs, expressed as defects per 100 units, shall be shown in table II.

TABLE II. Inspection levels and AQLs.

Examination	Inspection level	AQL (defects per 100 units)
4.4.1.1	S-4	2.5
4.4.1.2	S-2	4.0
4.6	S-2	4.0

4.4 Quality conformance inspection of the end item.

4.4.1 Examination of the end item. The end item shall be examined for the defects specified in 4.4.1.1 and 4.4.1.2 at the inspection levels and AQLs specified in 4.3. The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in glass ampoules of the specified capacity for the examinations specified in 4.4.1.1 and 4.4.1.2, and in units of shipping containers as specified in 4.6.

4.4.1.1 Examination of the product for appearance. The sample shall be one filled-unit container, and shall be examined as follows:

Examine	Defect
Appearance	Any foreign material; not clear, homogeneous liquid

4.4.1.2 Examination of the end item container for net contents. The sample unit for this examination shall be one glass ampoule. A minimum of five sample glass ampoules or the entire lot, whichever is smaller, shall be randomly selected from the lot and examined for volumetric contents. The average value shall be not less than the amount specified in the contract or purchase order (see 6.2).

4.4.2 Testing of the end item. The lot size shall be expressed in grams of 3-HEPT. The sample unit for testing shall be a sealed ampoule containing 10 milliliters of 3-HEPT. Sampling level for inspection shall be in accordance with level S-4 of MIL-STD-105, with an AQL of 2.5. Care shall be exercised to prevent contamination or alteration of the antiwetting compound 3-HEPT during storage and testing. The sample lot shall be unacceptable if the composite fails to conform to any of the requirements specified in tables I or III.

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TABLE III. Instructions for testing the end item.

Characteristic	Requirement	Test method	Requirement applicable to compound sample	Number determinations per unit	Results reported as
Silane content, (as 3-HEPT)	Table I	4.5.1	X	1	Determination per unit tested to nearest 0.1 percent
Distillation range at 35 mm	Table I	4.5.2	X	1	Pass or fail

4.5 Test methods.

4.5.1 Testing for silane content. The silane content of the sample shall be determined by gas-liquid phase chromatography using a flame ionization detector. The column packings used for this analysis shall be 10 percent SE30 or a combination column of 0.4 percent DV 7 and 1.5 to 1.95 percent SP 2250 and SP 2401. Column temperature shall be 125°C (257°F). ASTM E 260 and E 355 shall be used as a reference for this analysis.

4.5.2 Distillation range at 35 mm (1.38 inches) of Hg. ASTM E 133 shall be used as a reference for the equipment required for the distillation. Using state-of-the-art vacuum distillation techniques, the initial boiling point shall be 85°C (185°F) at 35 mm of Hg as indicated by the distillation thermometer at the instant the first drop of condensate leaves the condenser tube. The final temperature of the distillation shall be 87°C (188.6°F) at 35 mm (1.38 inches) of Hg. The final temperature is the temperature indicated at the instant the last drop of liquid evaporates from the lower point in the distillation flask or the highest temperature reached on the distillation thermometer before the distillation stops.

4.6 Inspection of packaging. Inspections shall be performed to ensure that packaging, packing, and marking conform to the requirements of section 5. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully packaged shall be examined for closure defects as specified in table IV.

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TABLE IV. Examination for closure defects.

Examination	Defect
Packaging	Not level specified Materials and construction not as specified Evidence of leakage Ampoules not scored completely around neck Not properly labeled
Packing	Not the specified level Materials and construction not as specified Any nonconforming component; component missing, damaged, or otherwise not serviceable Any bulged, distorted, or broken container
Count	Less than specified number of packages per shipping container
Markings (interior or exterior, as applicable)	Omitted, illegible, incorrect, incomplete; not in accordance with contract requirements.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Level A preservation. Antiwetting compound 3-HEPT shall be unit preserved in color break, constricted stem type ampoules of 10 mL capacity. The ampoules shall be manufactured from borosilicate conforming to type I glass as specified in the U.S. Pharmacopeia.

5.1.1 Intermediate packing. Five ampoules of the compound shall be packed in a paperboard set-up or folding carton. Ampoules shall be segregated in the carton by wrapping each ampoule in cushioning material or partitions that will prevent the ampoules from contact during handling and shipping.

5.2 Packing. Packing shall be level B or C as specified (see 6.2).

5.2.1 Level B. Antiwetting compound preserved as specified in 5.1, shall be packed in fiberboard boxes conforming to PPP-B-636 class weather-resistant. Boxes shall be closed method V in accordance with the appendix to the box specification.

5.2.2 Level C. Antiwetting compound shall be packed as specified in 5.2.1, except that boxes shall be of the domestic class and method I closure shall apply.

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5.3 Marking. In addition to any special marking required (see 6.2) and herein (see 5.3.1), interior packs (see 5.1.1) and exterior shipping containers shall be marked in accordance with MIL-STD-129. Bar code marking of MIL-STD-129, when specified (see 6.2) shall be on the intermediate pack and exterior shipping containers only.

5.3.1 Special marking.

5.3.1.1 Ampoules. Each ampoule shall be clearly marked with the following precautionary information:

"National Stock Number
Antiwetting Compound 3-HEPT
NOT FOR DRUG USE
Corrosive
Read Cautions on Box Label"

5.3.1.2 Intermediate pack. In addition to the marking specified in 5.3, each intermediate pack shall include the following precautionary marking:

"WARNING

Evolves corrosive and noxious hydrogen chloride fumes.
Material flammable prior to formation of fiber.
Do not breath vapors.
May cause burns.
In case of contact, immediately flush skin or eyes with plenty of water;
For eyes, flush for at least 15 minutes and get immediate medical attention.
STORE IN A COOL PLACE"

5.3.1.3 Hazardous labels. In addition to the markings above, each intermediate pack and shipping container shall contain the applicable hazard label specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The antiwetting compound 3-HEPT covered by this specification is intended for use in coating periscope headwindows.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of sample units to be tested, if other than specified (see 4.2).
(c) Amount of compound per container (see 4.4.1.2).
(d) Level of packing required (see 5.2).
(e) Special marking required (see 5.3).

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6.3 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets (MSDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313. In order to obtain the MSDS, FAR Clause 52.223-3 must be in the contract.

6.4 Subject term (key word) listing.

Antiwetting compound
Distillation range
3-HEPT
Silane

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6850-N833)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-28844A(SH)	ANTIWETTING COMPOUND 3-HEPT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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